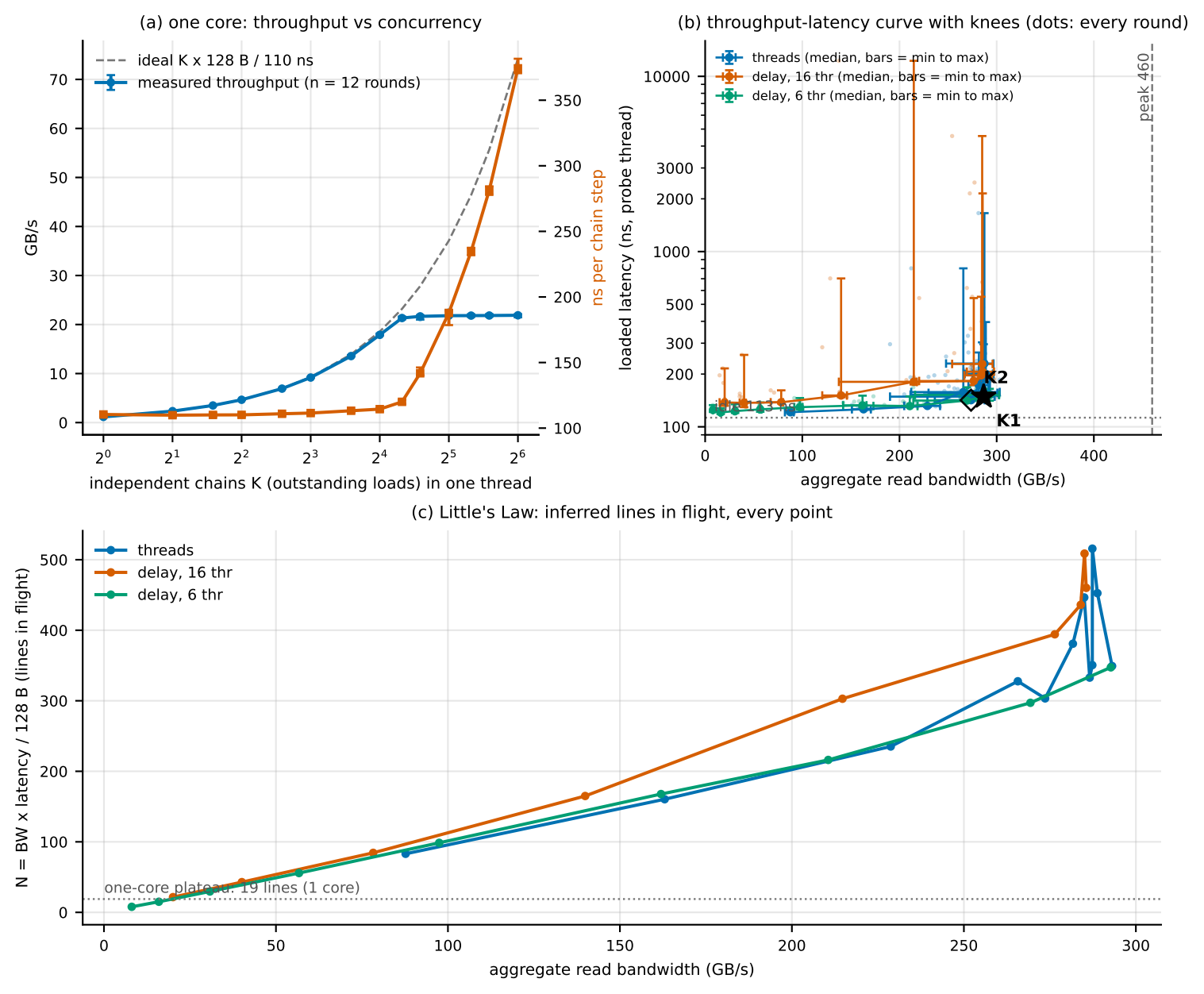
(a) one core: throughput vs concurrency
ideal K x 128 B / 110 ns
measured throughput (n = 12 rounds)
GB/s
ns per chain step
independent chains K (outstanding loads) in one thread
(b) throughput-latency curve with knees (dots: every round)
threads (median, bars = min to max)
delay, 16 thr (median, bars = min to max)
delay, 6 thr (median, bars = min to max)
peak 460
loaded latency (ns, probe thread)
K2
K1
aggregate read bandwidth (GB/s)
(c) Little's Law: inferred lines in flight, every point
threads
delay, 16 thr
delay, 6 thr
N = BW x latency / 128 B (lines in flight)
one-core plateau: 19 lines (1 core)
aggregate read bandwidth (GB/s)